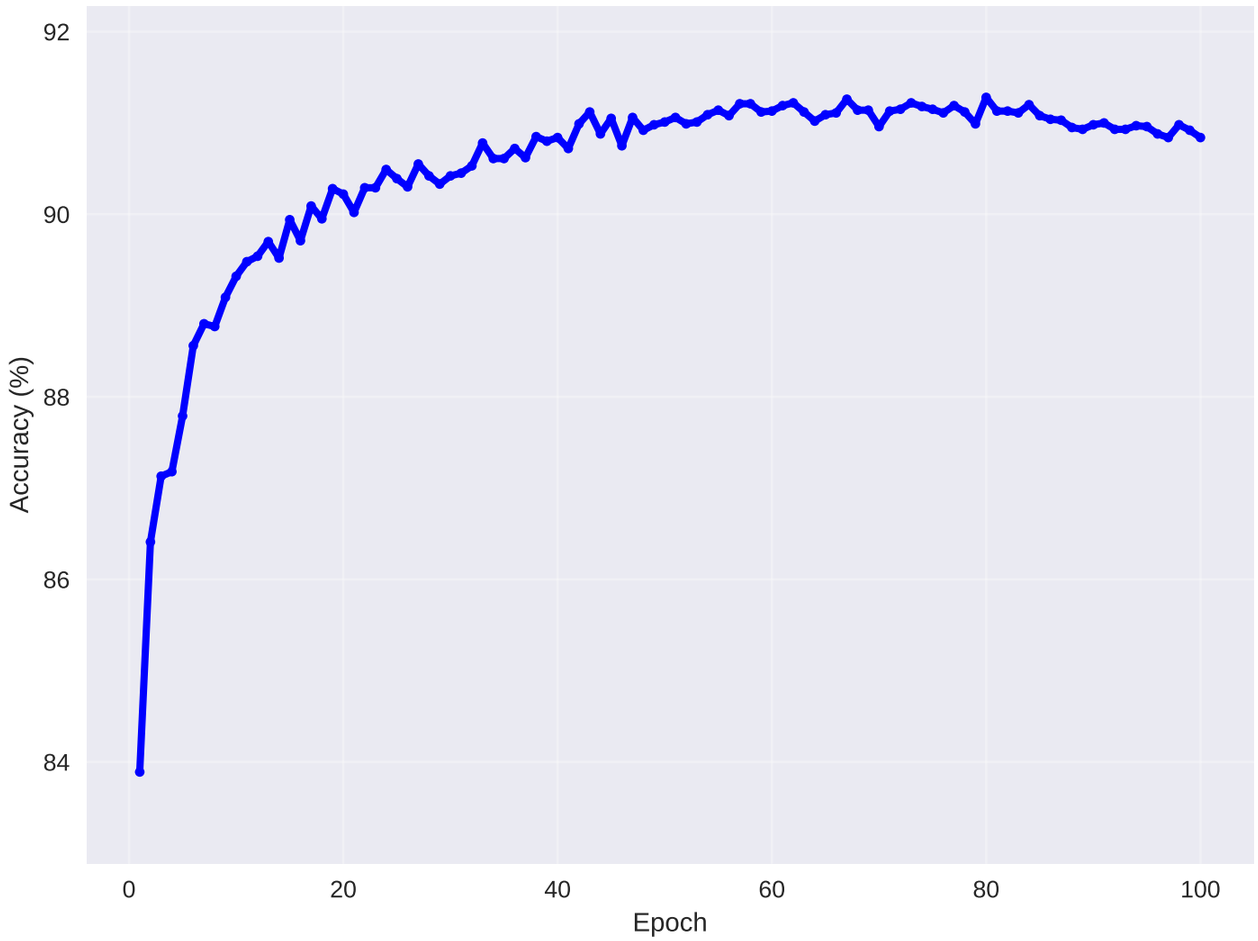
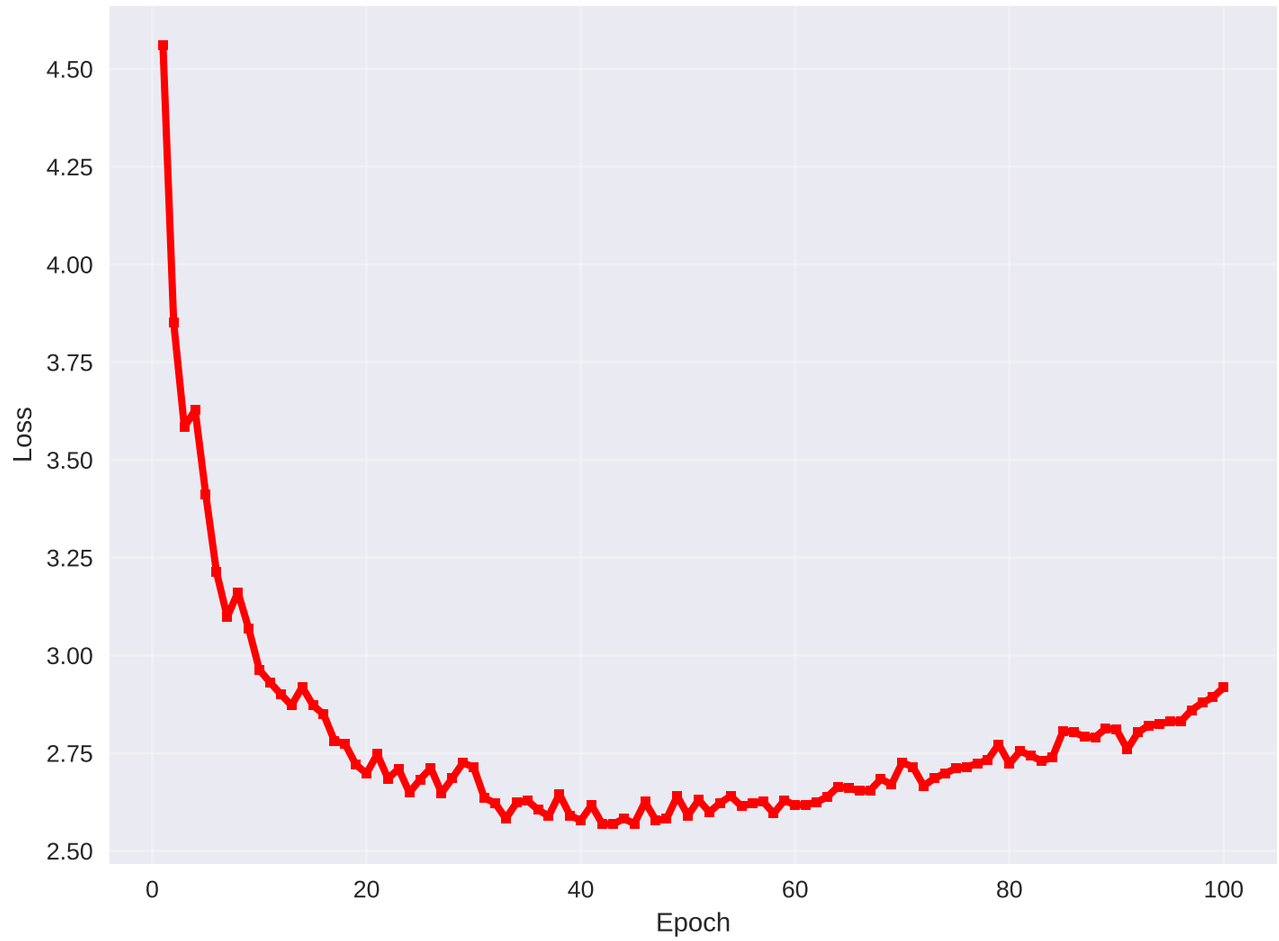
Federated Learning - Main Metrics Overview

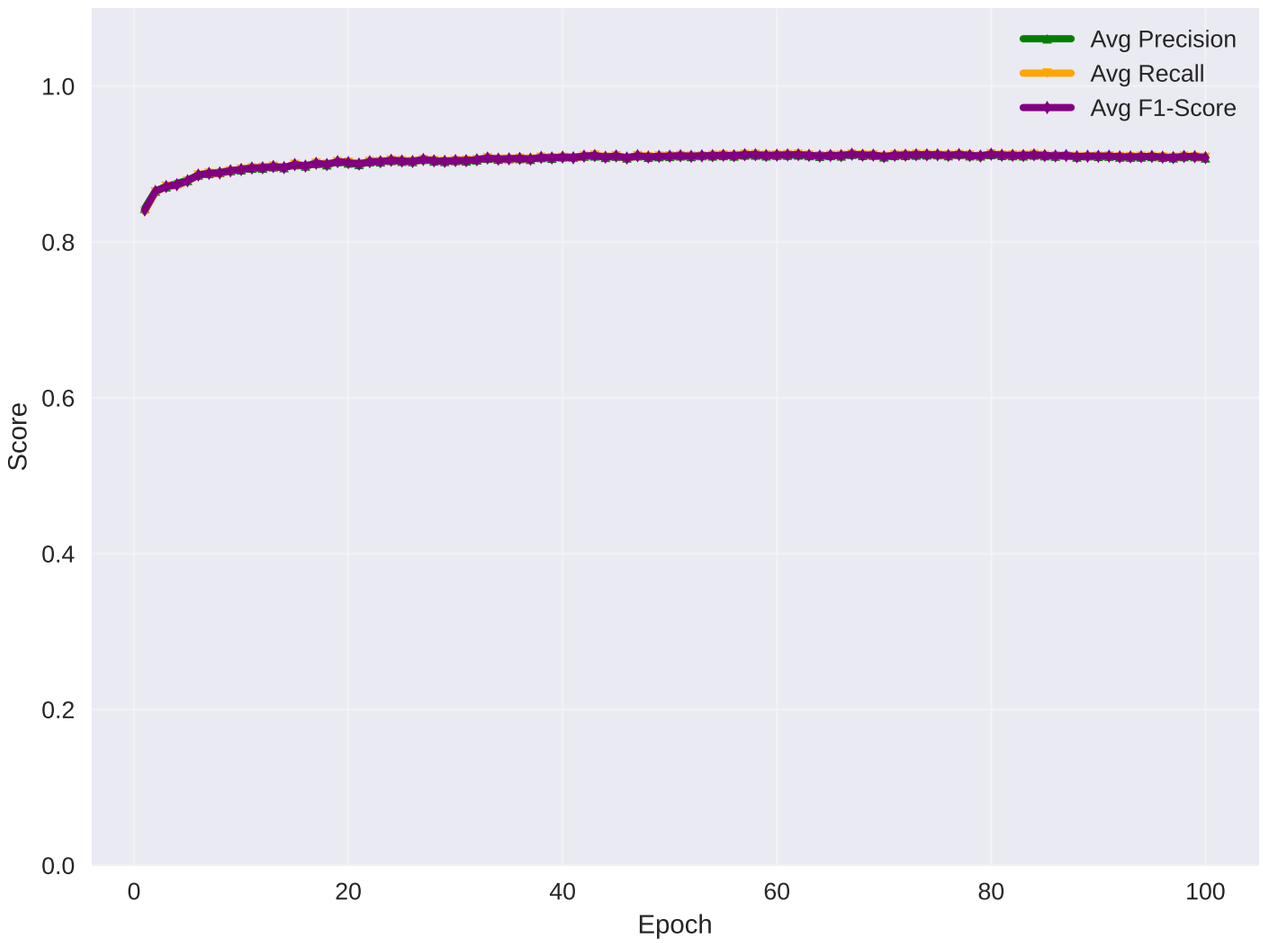
 Global Model Accuracy Evolution



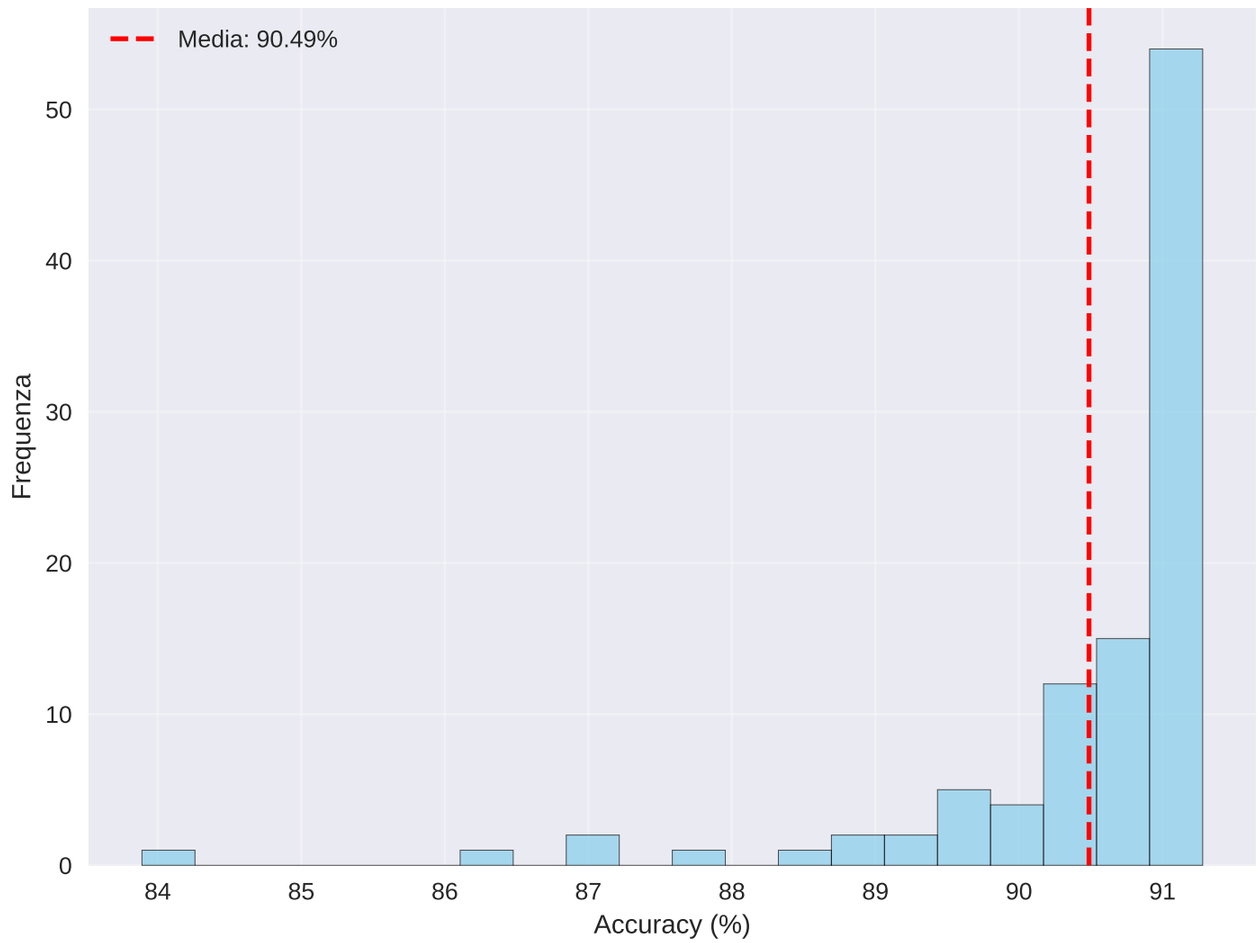
 Global Model Loss Evolution



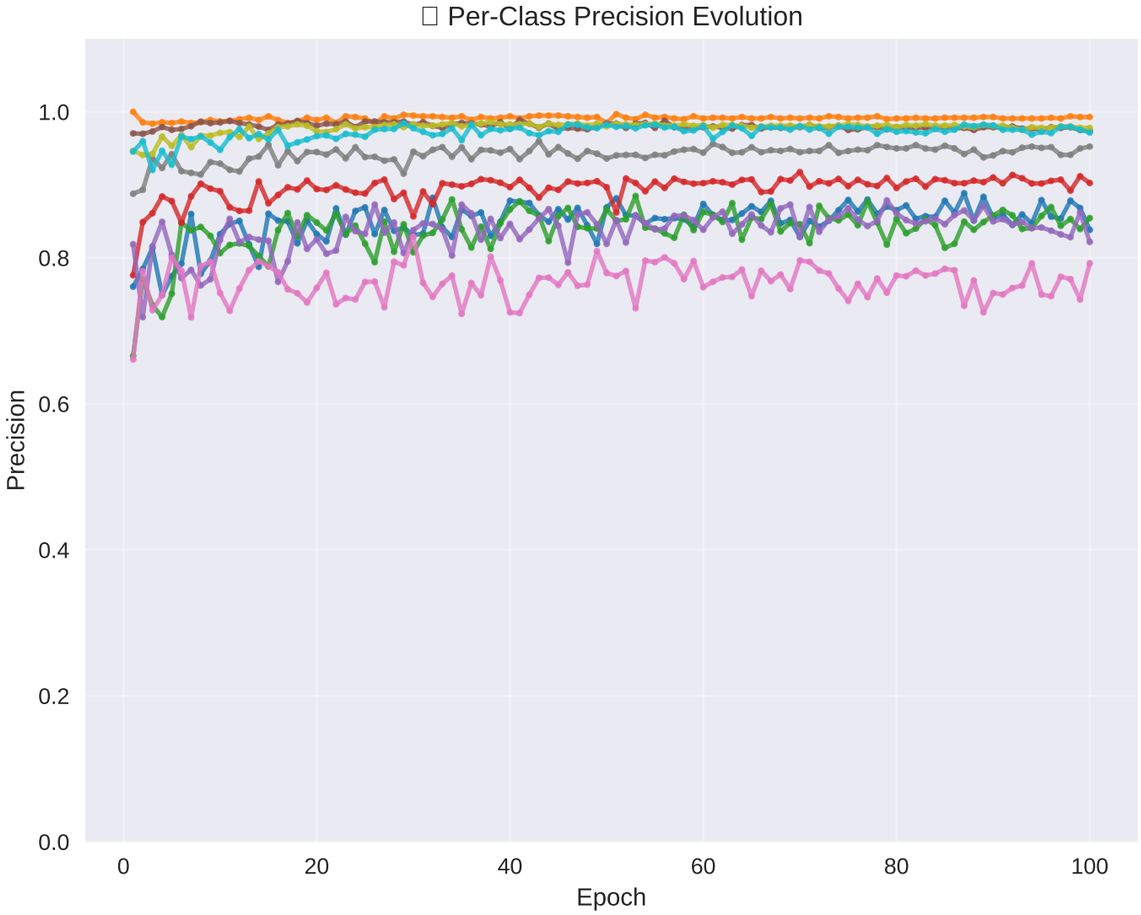
 Average Precision, Recall & F1-Score

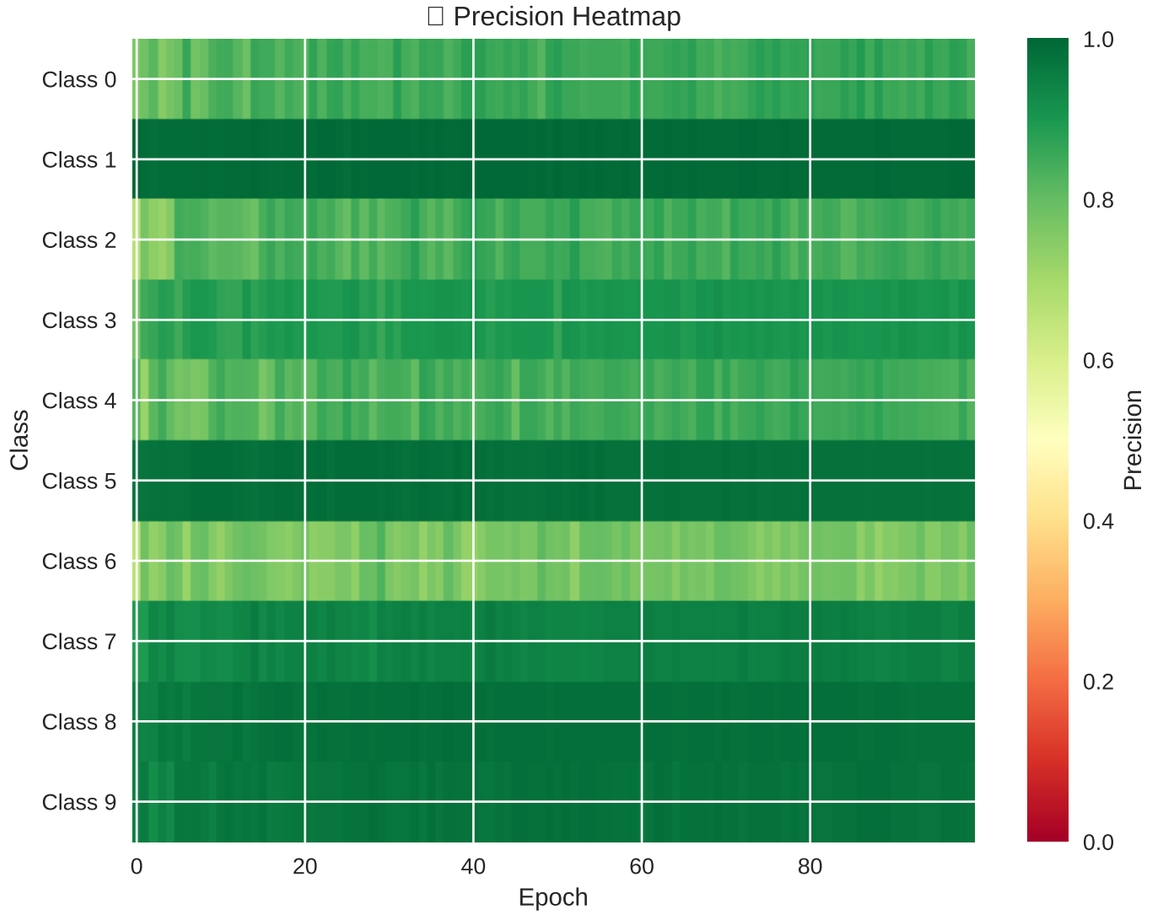


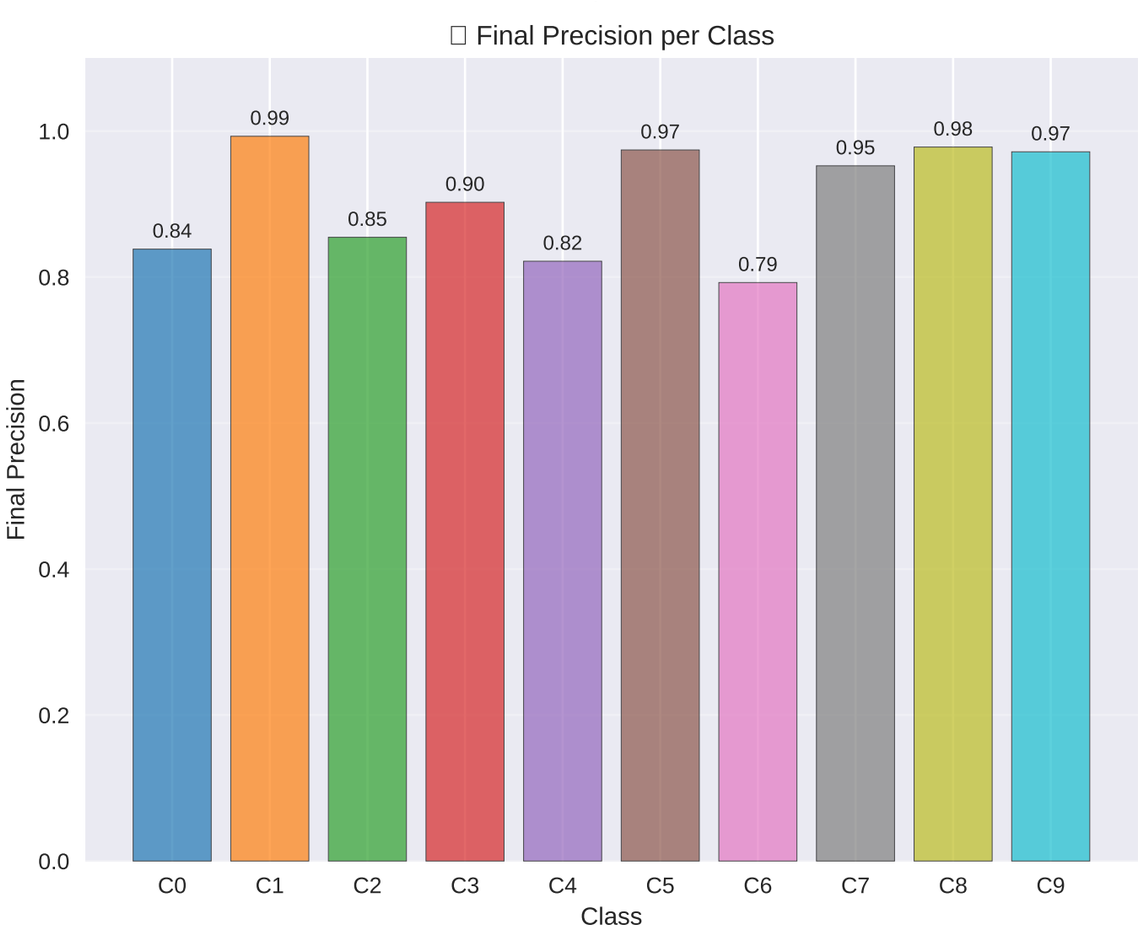
 Distribuzione Accuracy

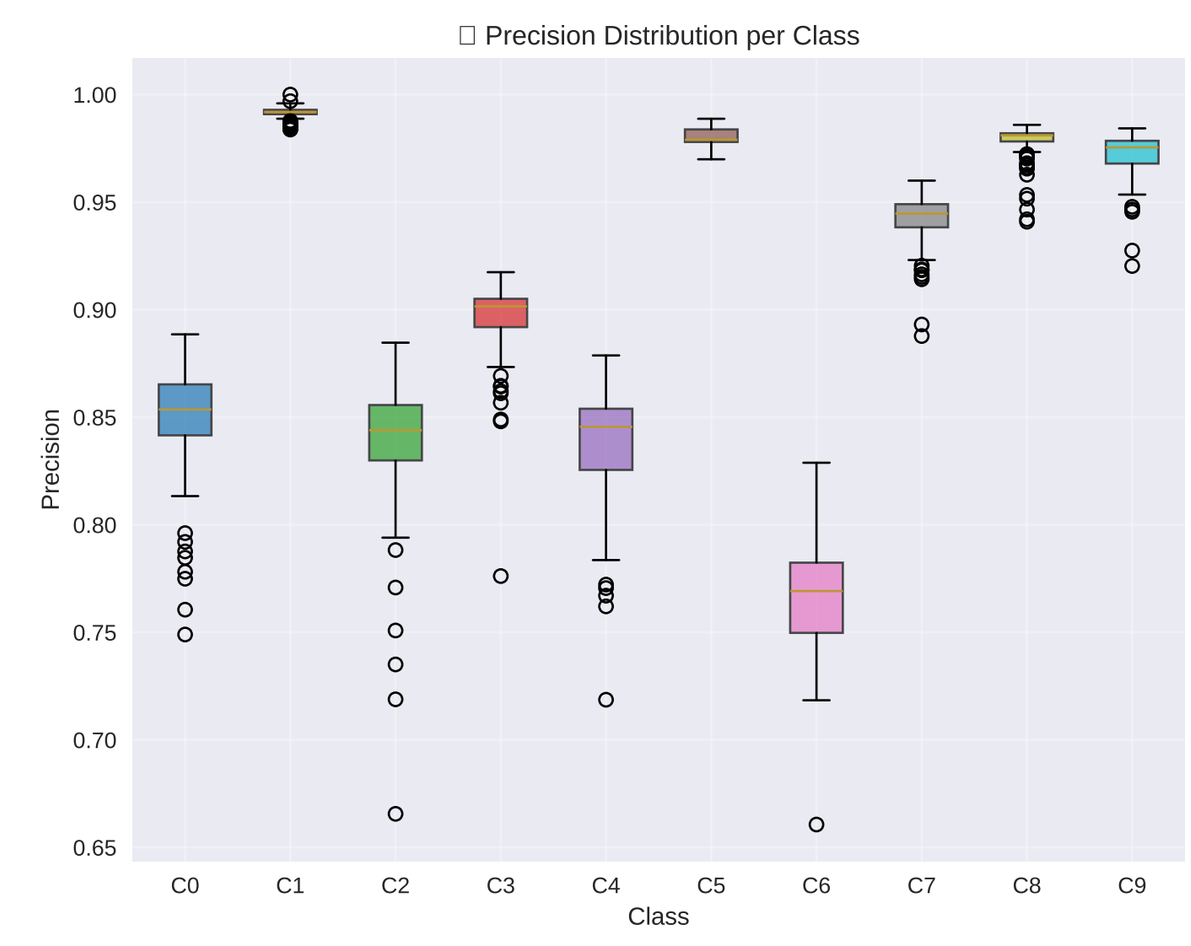


Per-Class Precision Analysis

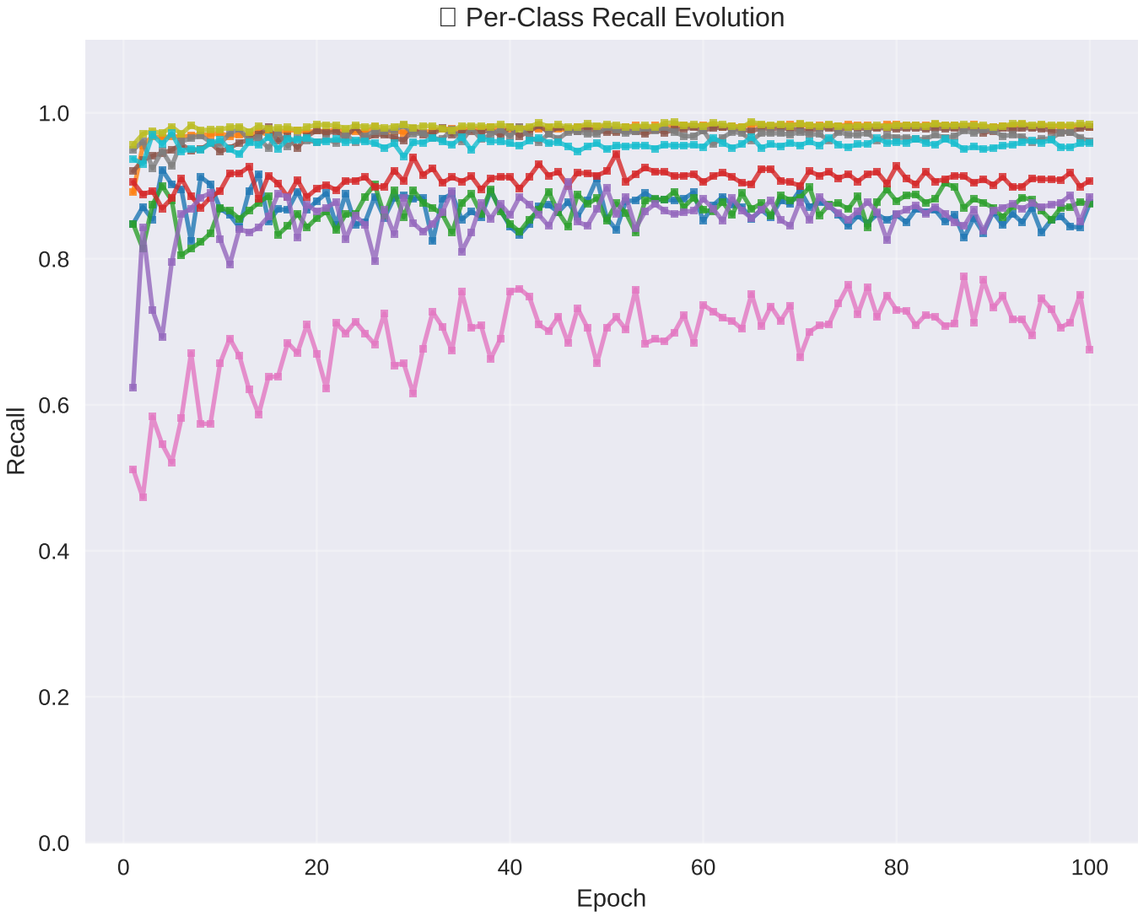


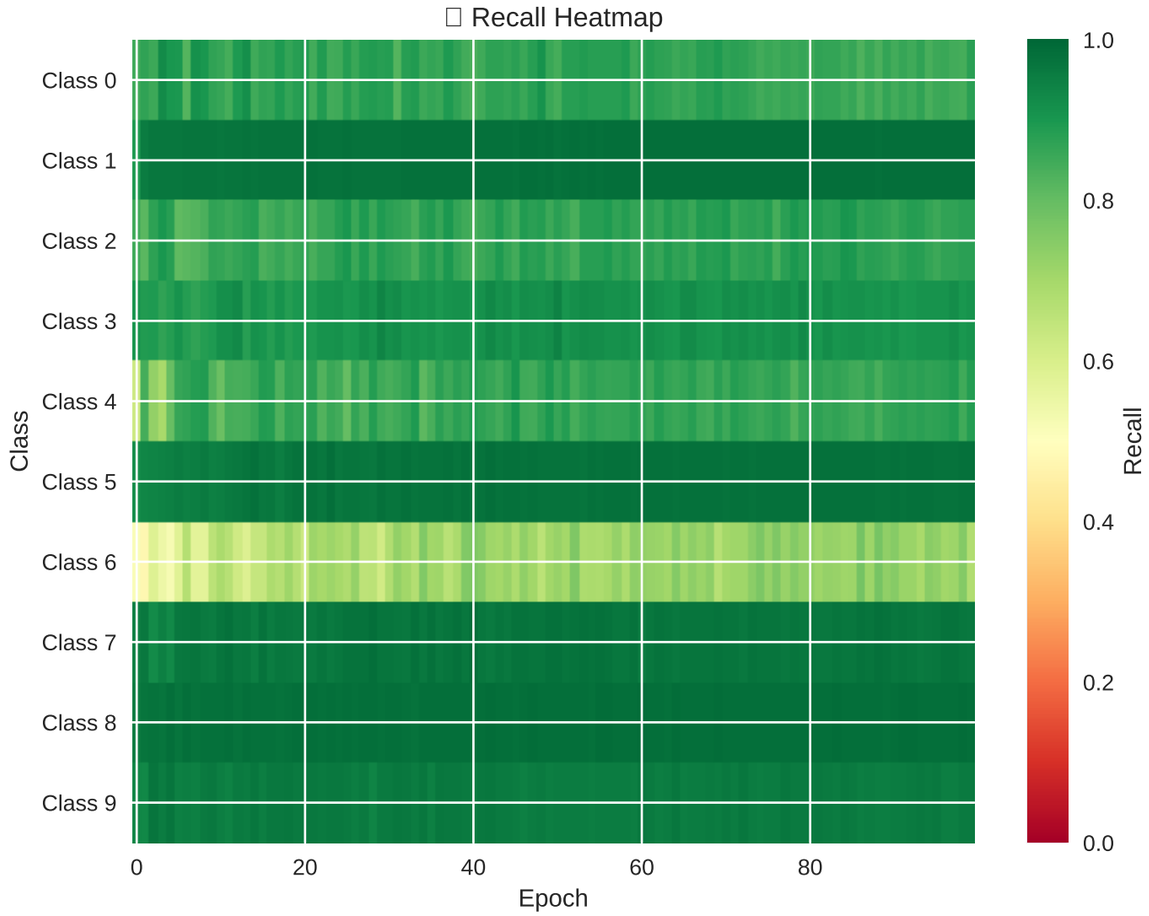


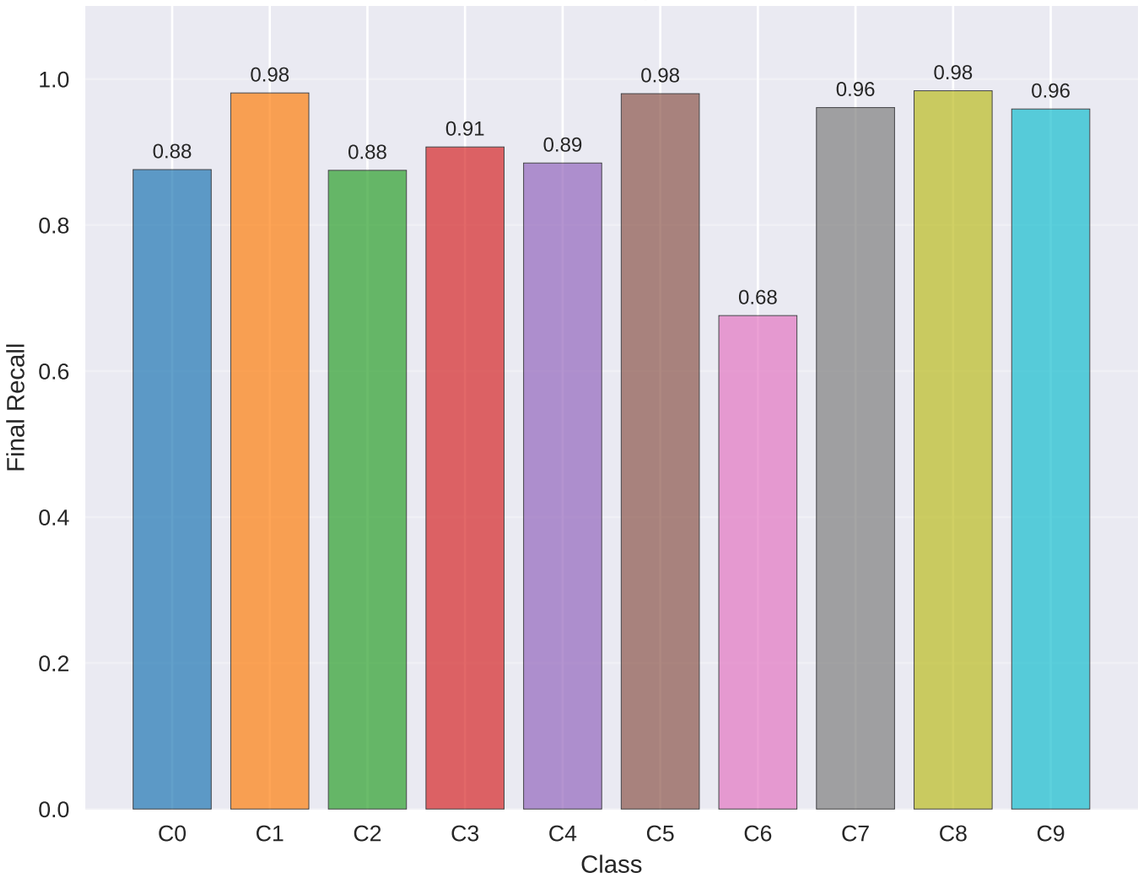


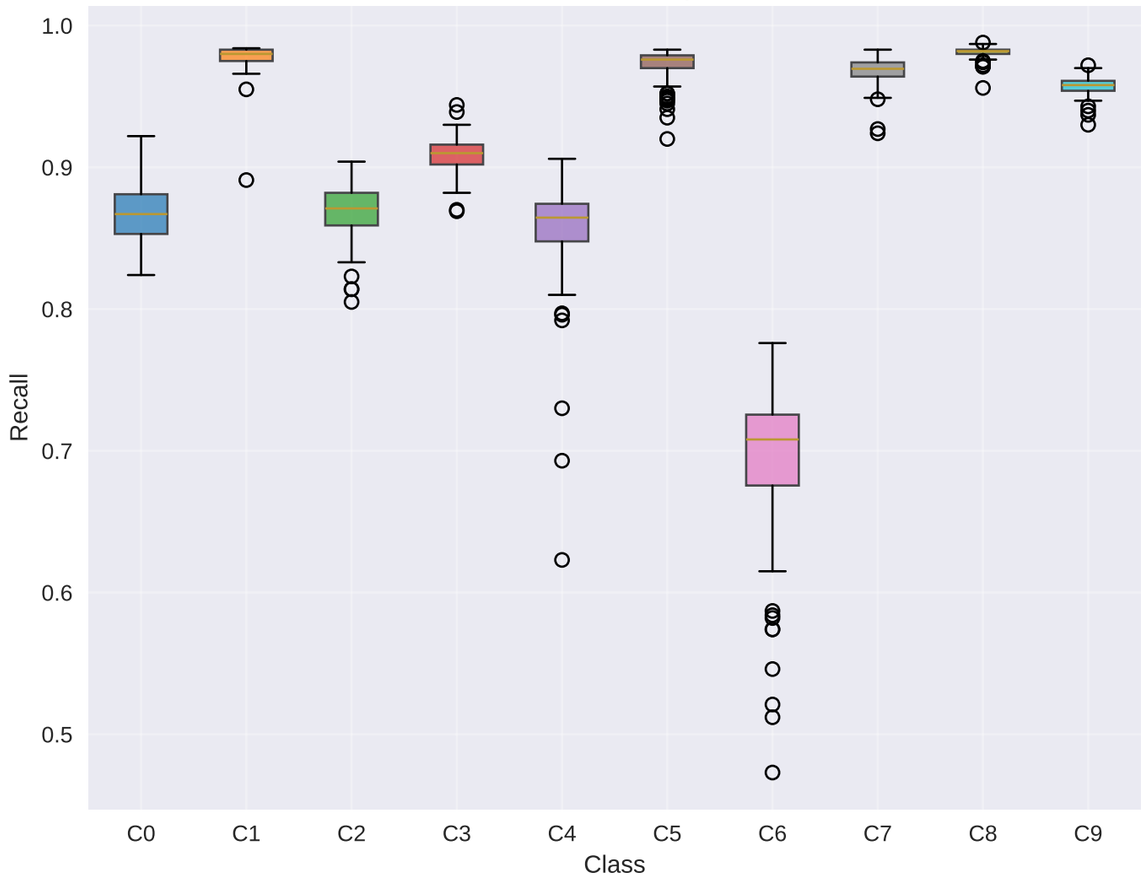


Per-Class Recall Analysis

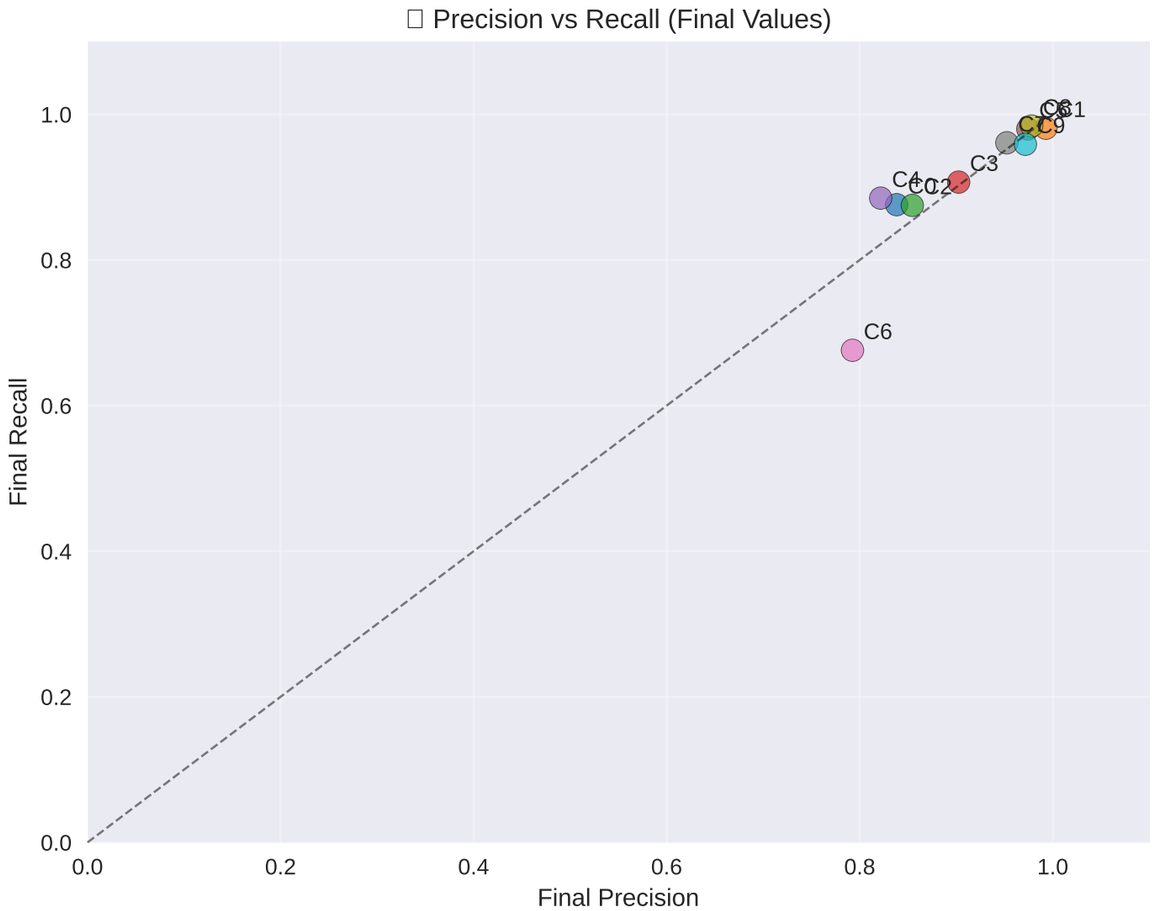




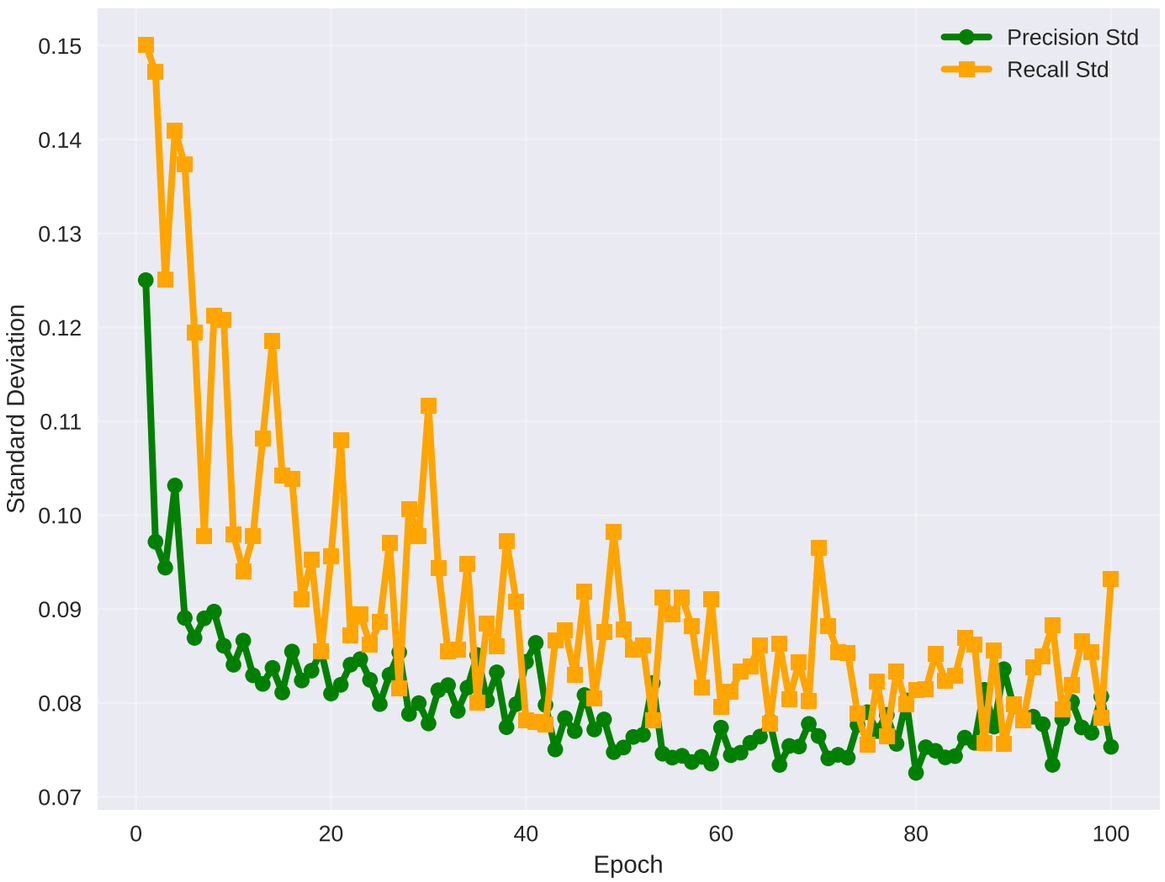


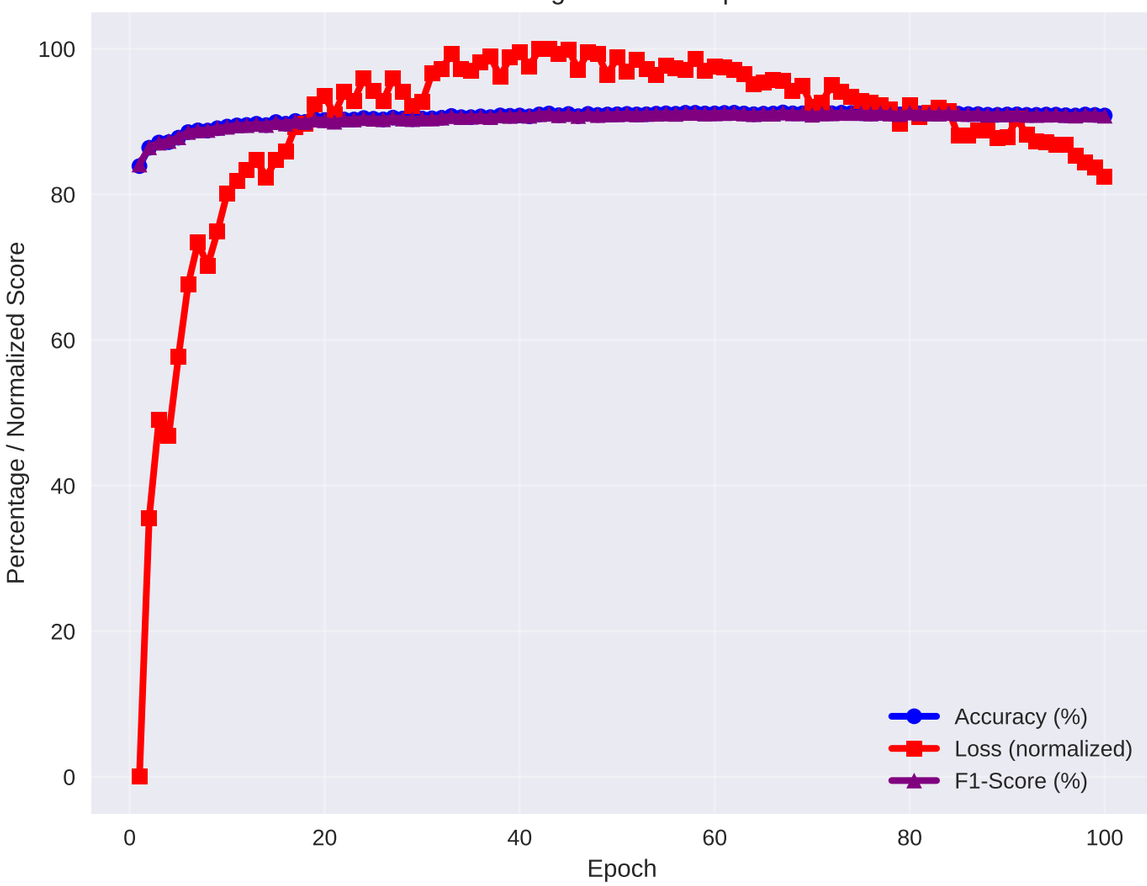


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-31 16:34:01
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 90.84%
• Loss: 2.9180
• Average Precision: 0.908
• Average Recall: 0.908
• Average F1-Score: 0.908

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.95%
• Riduzione Loss: +1.6430
• Stabilità Accuracy (CV): 0.0126

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.993)
• Peggiore Precision: Classe 6 (0.792)
• Migliore Recall: Classe 8 (0.984)
• Peggiore Recall: Classe 6 (0.676)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.838, R=0.876, F1=0.857
• Classe 1: P=0.993, R=0.981, F1=0.987
• Classe 2: P=0.854, R=0.875, F1=0.865
• Classe 3: P=0.902, R=0.907, F1=0.905
• Classe 4: P=0.822, R=0.885, F1=0.852
• Classe 5: P=0.974, R=0.980, F1=0.977
• Classe 6: P=0.792, R=0.676, F1=0.730
• Classe 7: P=0.952, R=0.961, F1=0.957
• Classe 8: P=0.978, R=0.984, F1=0.981
• Classe 9: P=0.972, R=0.959, F1=0.965

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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